

Here are some of the benefits offered by video analytics technology:

Effective management of real-time CCTV video. Video analytics allows you to effectively manage multiple CCTV cameras within a CCTV system. This software watches the cameras for you and notifies you when action needs to be taken. This requires fewer security personnel to watch more cameras than a standard CCTV system.

Intelligent motion detection. Video analytics software is based on various algorithms that will analyse the video to determine the type of motion. This means your CCTV system will only look for the true motion of a person or vehicle.

Number plate recognition. This system allows the camera to detect automobile licence plates that can then be integrated in the system for behaviour monitoring. It also allows cameras within the CCTV system to determine the direction the vehicles are heading towards.

Object recognition. Video analytics within a CCTV system also allows one to monitor for missing or unwanted objects.

Auto-tracking motion with PTZ cameras in a CCTV system. Video analytics can also be coupled with pan/tilt/zoom (PTZ) cameras within a CCTV system, which enables auto-tracking capabilities or real-time tracking.

While the industry is still exploring and weighing its potential, more functions and features are being introduced in these systems. These include smoke detection, behavioural analysis, features for military applications, forensic studies and 3D volumetric analysis.

What's latest in the market?

Video analytics systems can be used in any camera and customised to run on any specification.

Says Jindal, "Many embedded analytic algorithms can compensate for various camera issues and quality levels, making possible the use of lower-cost cameras for achieving better range and resolution. Also, a recent trend towards tighter integration with adjacent systems



Bosch IVA raising an alarm

including video management, physical security information management, point-of-sale and building automation, offers broader integration and interoperability among various verticals."

Surveillance Site IQ from Siemens is an automated wide-area video analytics software, which is capable of automated surveillance, and has features like intelligent alarm monitoring and interactive event responses through the analysis of data from cameras and sensors. Widely used for perimeter intrusion detection, this system can classify an object.

Informs Sundaram, "This software requires only 10 pixel changes to classify an object and hence a fewer number of cameras are required. It provides the output on a single screen after collecting and analysing the data from hundreds of cameras and sensors. The data thus collected is represented on a 3D image of the entire site."

The single-screen philosophy adopted by Siemens helps in utilising

the security personnel effectively.

Axis Communications recently introduced a new chip, the ARTPEC-4, which enhances the image quality and video analytics performance in Axis' network video cameras and video encoders. It is designed to offer video with lower noise and higher light sensitivity for sharper images of moving objects. With a powerful CPU and a co-processor for accelerating video analytics, ARTPEC-4 also has more processing power for intelligent video analysis.

Contrary to the traditional solutions that use a central server to analyse video content, Bosch video analytics is embedded in cameras, making it convenient and efficient. Bosch IVA performs multi-level image analysis of pixels, textures and motion content, from inside the camera.

Shares Sahney, "Using distributed intelligence, Bosch IVA creates a far more robust system. It has no central server and therefore no single point of failure. The IVA operates independently in each component—even if one camera or encoder fails, the rest of the system remains fully functional." ●

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The author is assistant editor at EFY. The article is reproduced from Electronics Bazaar magazine of EFY Group